

GRADE 9 MATHEMATICS • MEASUREMENT & GEOMETRY

Circle Geometry

THE BIG IDEA

Every circle problem comes from a few key properties — and the inscribed angle theorem is the one you'll use most.

INSCRIBED ANGLE THEOREM

inscribed angle = $\frac{1}{2} \times$ central angle

(same arc)

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al angle

WORKED EXAMPLE

Central angle $AOB = 110^\circ$

Inscribed angle $APB = 110^\circ \div 2 = 55^\circ$

Major arc $= 360^\circ - 110^\circ = 250^\circ$

Inscribed angle AQB (major arc) $= 250^\circ \div$

⚠ WATCH OUT FOR

MISTAKE

"All inscribed angles equal 90° "

MISTAKE

"Inscribed angle = central angle"

MISTAKE

$$2 = 125^\circ$$

FIX

Only ones that subtend a **diameter**

FIX

Inscribed angle is **half** the central angle

FIX

Forgetting the right angle at the tangent point



SELF-CHECK

If a central angle is 84° , what is the inscribed

Tile set of 6 views 30 cm x 30 cm
Circle geometry poster — tape to matching edges

**Tangent $\perp$ radius — mark it before using
Pythagorean theorem**

angle on the minor arc? On the major arc?